

Latent Enclosures, Loudspeakers, Microphones, and Custom Acoustics Analysis System (2025)

Latent Enclosures is a site-adaptable sound installation developed and presented at the 2025 Ars Electronica Festival that explores how machine listening systems perceive, transform, and rearticulate architectural acoustics.

Using impulse response analysis and real-time resynthesis, the work reveals latent sonic properties of enclosed spaces and allows listeners' movements within the installation to dynamically alter its acoustic behavior.

Detailed Information about the work:

Latent Enclosures is a sound installation that reveals, transforms, and resynthesizes latent sonic phenomena within an enclosed space. A custom sound analysis system captures impulse responses from multiple locations within a room and processes them to uncover acoustic qualities perceptible and imperceptible to the unaided human ear: prominent and non-prominent resonances, noisiness, temporal reflections, and other emergent characteristics detected by automated machine listening algorithms.

Recorded sounds from within the installation and real-time acoustic feedback are filtered through these transformed impulse responses, which articulate and modulate the room's latent acoustic properties. As listeners move among strategically placed microphones and speakers, their physical presence alters the room's acoustics, and their auditory experience fluctuates in relation to their position within the space.

The same analytical processes applied to the room's impulse responses are also used to deconstruct recordings captured within the installation. This resynthesizes the machine's perception of these sonic events while establishing a link between sounds produced by participants, environmental sounds originating beyond the installation space, and the ways in which obfuscated, enclosed digital systems render sound and shape auditory experience.